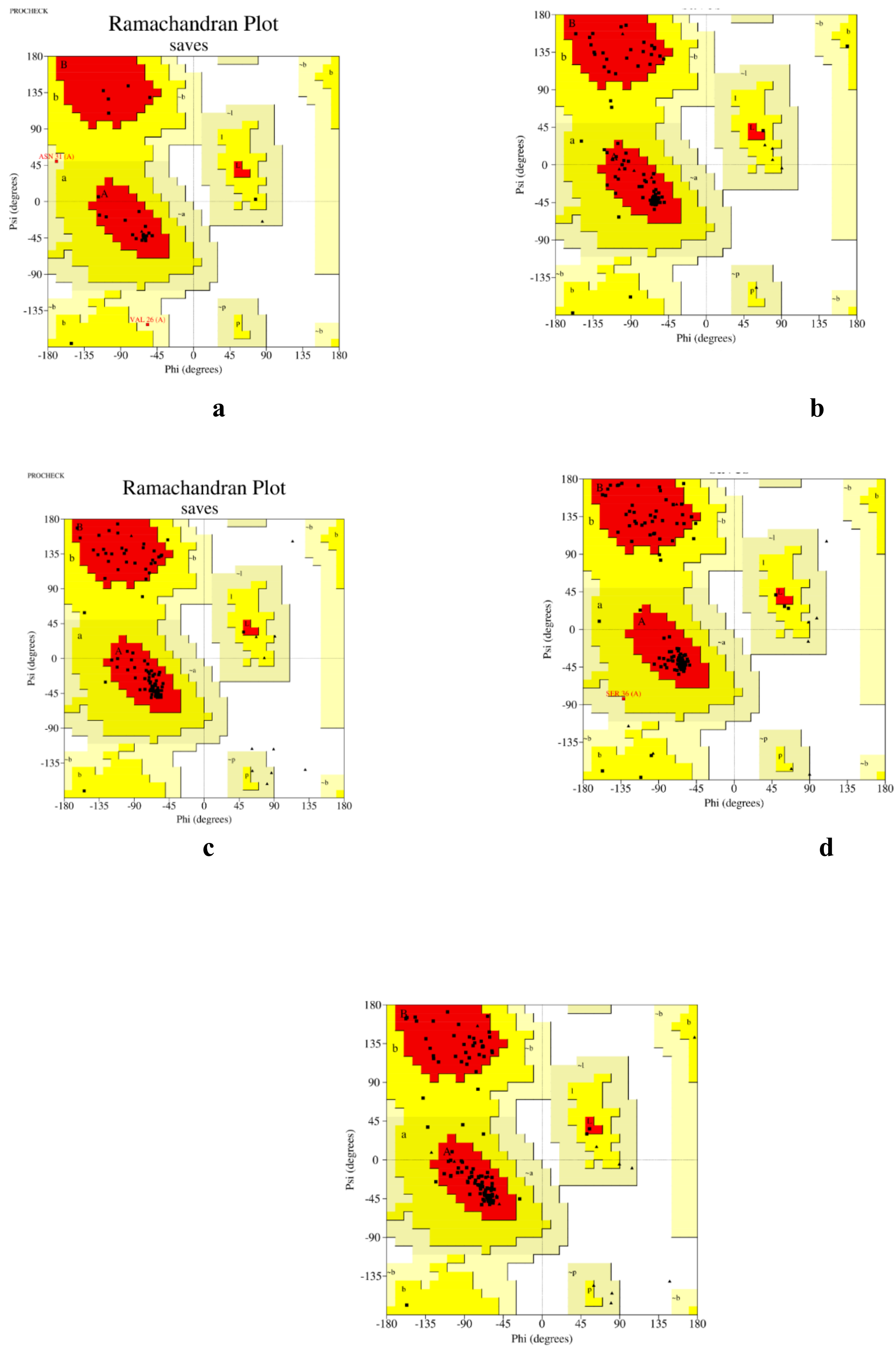


Figure 1: Ramachandran plots for models **a** P0DKT6, **b** Q4Q03, **c** P00620, d P21789, and **e** B5U6Z2.

Table 1: Residue names and numbers of surface binding pockets considered to guide docking.

Protein		**Active residues used**
Snake Venom PLA2s	P0DKT6	Lue2, Ile3, Glu4, Phe5, Gly6, Lys7, Met8, Thr10, Glu11, Glu12, Pro16, Val17, Phe18, Pro19, Tyr20, Glu21, Ala22
	Q4QT03	Val18, Phe21, Ile25, Pro33, Tyr37, Tyr43, Cys44, Gly47, Gly48, Cys60, His63, Asp64, Tyr67, Lys76, Phe112
	P00620	Leu2, Phe5, Gly6, Ile9, Val16, Tyr19, Ile20, Tyr25, Cys26, Gly27, Cys42, His45, Asp46, Tyr49, Lys58, Phe93
	P21789	Lue2, Phe5, Gly6, Ile9, Pro17, Tyr21, Gly22, Tyr27, Cys28, Gly29, Cys44, His47, Asp48, Ala92, Phe96
	B5U6Z2	Val18, Phe21, Gly22, Ile25, Pro33, Phe34, Tyr37, Thr38, Tyr43, Cys44, Leu46, Cys60, His63, Asp64, Tyr67, Asp105, Phe112
***Mtb* membrane protein**	4QLP	Phe100, Gln101, Glu102, Leu105, Ile111, Thr113, Arg119, Gln120, Arg121, Asp122, Ala136, Ala137, Asp138, Val141, Trp798, Pro800, Glu801, Gly802, Trp803, Asp827, His828, Gln829, Arg841
	7WCJ	Pro40, Ala41, Thr42, Asp43, Gly44, Ala45, Thr46, Ala49, Arg53, Leu69, Arg71, Pro73, Asn74, Gln76, Arg77, Leu78, Ala95, Asp97, Val98, Val99, Trp100, Thr149, Thr150, Asn151, Gln153, Trp187, Ala189, Val190, Gln191, Ala192, Asn193, Gln194, Gly195, Glu196, Ala197, Leu198, Val200, trp201, asp250, ser252, ile253, thr256, glu257, glu258, gly259, ser260, ala261, arg262, leu263, ala264, glu266, gln267, val274, Asn275, Trp276, Pro277, Phe278, Phe280, Ala281, Ser282, Leu284, Glu285, Asn286, Gly291, Val292, Asn308, Ile310, Thr312, Phe313, Thr314, Pro315, Ser316, Asp317, Phe320, Tyr336, Gly350, Gly351, Leu352, His377, Asn380, Tyr381, Ser383, Leu384, Glu385, Gly386, Gly387, Leu388, Pro389, Gln400, Ala403, Lys404, Tyr405, Pro406, Met407, Arg421, Tyr427, Gln428, Ala431, Val432, Ala435, Ser439, Pro440, Ile441
	7NVH	Thr39, Gln40, Asp64, Ser66, Gly67, Val70, Ala164, Gly165, Leu166, Val169, Ala170, Leu173, Thr174, Ile177, His234, Tyr235, Phe236, Ile422, Ser423, Glu424, Gln437, Phe440, Tyr447, Thr448, Ser449, Leu693, Met695
	7UQ0	Asp64, Pro65, Gly66, Pro67, Arg70, Pro71, Ala72, His73, Asn74, Ala75, Ala76, Gln161, Gly162, Val178, Phe179, Thr180, Ala181,

		Ala182, Asp186, Asn189, Asn190, Arg192, Gly193, Ala196, Ala199, Ile201, Asp205, Ile208, Thr209
	6DDN	Ser144, Asp163, Pro176, Ala182, Lys183, Leu186, Leu187, Ser233, Tyr240, Ala270, Trp332, Asp336, Leu339, Phe340, Asp341

<u>Table 2:</u> HADDOCK docking with P0DKT6. Showing scores for the top-ranked cluster when docked to each of the Mtb membrane proteins.

Parameters	**Mtb membrane protein name**				
	4QLP	**7WCJ**	**7NVH**	**7UQ0**	**6DDN**
HADDOCK score	-109.9±4.6	-73.7±4.6	-88.7±55	-89.0±2.9	-72.5±13.8
Cluster size	32	16	44	78	7
RMSD from the overall lowest-energy structure	0.3±0.0.2	0.3±0.2	0.9±0.7	0.9±0.6	0.7±0.4
vdW Energy	-61.0±8.1	-62.5±6.8	-50.7±5.9	-61.6±4.2	-43.8±4.0
Electrostatic Energy	-219.9±28.2	-284.7±30.0	-223.4±28.4	-67.3±5.5	-105.0±34.1
Desolvation Energy	-13.5±2.0	-10.8±2.5	-17.7±2.0	-27.0±1.6	-13.0±1.4
Restraints Violation Energy	84.7±10.1	565.5±48.4	243.8±54.5	130.4±32.2	53.1±29.1
Buried Surface Area	2071.4±71.8	2350.5±46.7	2000.5±82.8	1715.2±41.3	1420.7±59.9
Z-Score	-2.5	-1.2	-2.1	-0.9	-1.5

vdW-van der Waals Energy.

<u>Table 3:</u> HADDOCK docking with Q4QT03. Showing scores for the top-ranked cluster when docked to each of the *Mtb* membrane proteins.

Parameters	***Mtb* membrane protein name**				
	4QLP	**7WCJ**	**7NVH**	**7UQ0**	**6DDN**
HADDOCK score	-99.2 ± 4.3	-66.2±8.9	-93.6±5.1	-93.2±1.9	-72.7±9.2
Cluster size	30	26	21	46	9
RMSD from the overall lowest-energy structure	0.7 ± 0.4	0.4±0.3	16.5±0.1	10.4±0.1	2.4±0.7
vdW Energy	-51.3 ± 3.7	-65.6±4.9	-77.5±4.2	-58.4±3.6	-37.8±4.6
Electrostatic Energy	-176.1 ± 7.2	-465.7±57.2	-210.3±24.0	-160.7±23.9	-147.8±36.0
Desolvation Energy	-22.9 ± 2.6	22.4±4.9	-12.1±3.1	-15.8±1.3	-19.4±0.9
Restraints Violation Energy	102.2 ± 14.2	701.9±97.0	380.4±32.4	131.3±21.5	140.0±22.6
Buried Surface Area	1954.1 ± 39.1	2749.9±78.6	2691.1±56.5	1857.9±43.2	1631.7±200.1
Z-Score	-1.7	-1.5	-2.3	-2.2	-1.4

Table 4: HADDOCK docking with P00620. Showing scores for the top-ranked cluster when docked to each of the *Mtb* membrane proteins.

Parameters	*Mtb* membrane protein name				
	4QLP	**7WCJ**	**7NVH**	**7UQ0**	**6DDN**
HADDOCK score	-143.6±2.6	-104.0±4.2	-117.6±3.8	-123.2±1.0	-115.3±8.3
Cluster size	127	39	4	132	7
RMSD from the overall lowest-energy structure	11.3±0.0	10.9 ±0.3	7.8±0.1	7.0±0.3	0.6±0.4
vdW energy	86.7±6.0	-81.8±3.8	-75.0±5.2	-61.7±5.3	-54.2±8.6
Electrostatic Energy	253.5±24.9	-228.1±28.3	-182.5+/-20.9	-172.8±16.7	-191.5±38.6
Desolvation Energy	-18.3±2.3	-35.5±4.2	-33.5±4.0	-35.2±2.9	-30.2±3.2
Restraints Violation Energy	121.2±48.9	589.8±52.9	273.5±83.6	82.4±49.6	74.1±16.0
Buried Surface Area	2505.8±61.9	2501.4±149.6	2512.2±30.4	1834.9±45.2	1848.4±76.8
Z-Score	-1.0	-1.4	-2.1	-1.3	2.4

Table 5: HADDOCK docking with P21789. Showing scores for the top-ranked clusters with each *Mtb* membrane protein.

Parameters	*Mtb* membrane protein name				
	4QLP	**7WCJ**	**7NVH**	**7UQ0**	**6DDN**
HADDOCK score	-121.9±5.5	-81.7±7.8	-120.9±6.5	-134.1±3.2	-125.3±10.7
Cluster size	7	40	9	36	16
RMSD from the overall lowest-energy structure	12.0±0.3	8.5±0.3	14.6±0.1	0.8±0.5	0.4±0.3
vdW Energy	-85.9±12.2	-79.1±5.8	-75.4±3.0	-61.5±4.6	-39.8±5.6
Electrostatic Energy	-201.0±45.8	-386.5±18.2	-229.2±29.3	-213.1±13.4	-500.6±38.5.6
Desolvation Energy	-32.4±6.8	8.8±2.7	-28.7±2.3	-39.2±4.4	-4.5±0.9
Restraints Violation Energy	366.6±55.1	659.0±74.5	291.4±39.1	91.9±19.0	192.2±27.6
Buried Surface Area	2778.3±147.1	2991.8±151.2	2447.5±101.1	2069.1±30.4	1948.7±59.1
Z-Score	-1.8	-1.9	-1.8	-1.4	-2.1

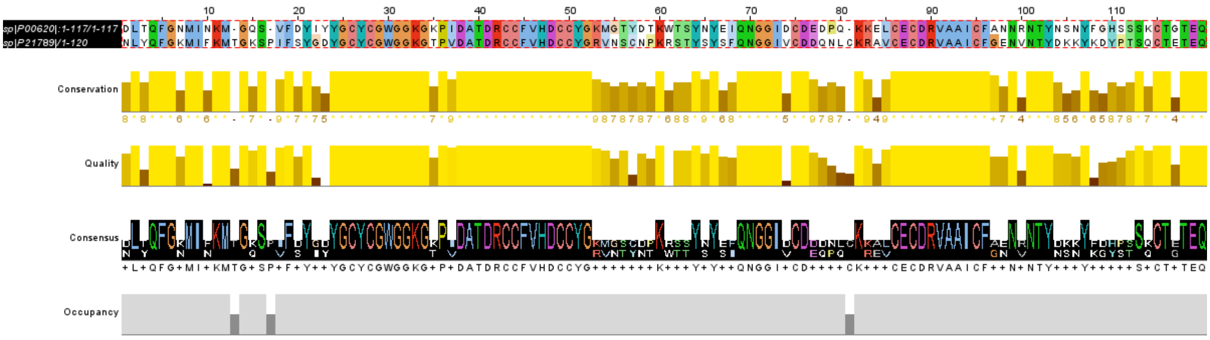
Table 6: HADDOCK docking with B5U6Z2. Showing scores for the top-ranked clusters with each *Mtb* membrane protein.

Parameters	**Mtb membrane protein name**				
	4QLP	**7WCJ**	**7NVH**	**7UQ0**	**6DDN**
HADDOCK score	-100.3±3.4	-60.1±6.9	-81.9±1.6	-130.2±2.5	-75.2±7.0
Cluster size	53	65	11	58	34
RMSD from the overall lowest-energy structure	13.3±0.1	0.6±0.4	17.2±0.1	0.5±0.4	4.6±0.3
vdW Energy	-58.2±1.9	-83.2±2.5	-58.4±4.3	-75.6±5.1	-59.7±8.8
Electrostatic Energy	-154.9±10.4	-222.5±25.1	-317.0±17.6	-180.6±35.5	-190.7±44.2
Desolvation Energy	-23.2±2.0	-8.4±4.1	3.6±3.0	-30.3±2.7	-5.9±3.3
Restraints Violation Energy	119.7±23.5	759.8±31.7	364.3±40.6	117.0±17.5	285.1±54.0
Buried Surface Area	1930±30.6	3023.5±137.9	2375.5±87.3	2402.7±43.5	2232.9±138.6
Z-Score	-1.9	-1.9	-1.9	-1.9	-1.8

```
Top 5 PLA2-Mtb protein complexes by combined interaction ranking:
      PLA2 Mtb_protein  Composite_score HADDOCK_score  Composite_rank
16  P21789        7WCJ             50.5         -81,7             1.0
6   Q4QT03        7WCJ             50.0         -66,2             2.0
21  B5U6Z2        7WCJ             46.0         -60,1             3.0
15  P21789        4QLP             39.2        -121,9             4.0
10  P00620        4QLP             36.5        -143,6             5.0
Kruskal-Wallis test for Composite_score differences between Mtb proteins: H=7.237, p=1.239e-01
```

Figure 2: A screenshot of the top ranks of interaction analysis of HADDOCK generated svPLA2-*Mtb* membrane

The most suitable template identified for the svPLA2s, P00620 and P21789, was 1VIP (anticoagulant class II phospholipase A2 from the venom of *Vipera russelli russelli*). Although both templates had an E-value of 0.0, their sequence identities to the template differed substantially—59% for P00620 and 77% for P21789. A pairwise sequence alignment was performed and is shown in the figure below.



Pairwise sequence alignment of P21789 (PA2A_CERCE) and P00620 (PA2A_BITGA) using EMBOSS NEEDLE Pairwise Sequence Alignment (PSA) via EMBL-EBI webserver (https://www.ebi.ac.uk/jdispatcher/psa/emboss_needle). The matrix used is the EBLOSUM62. The alignment length was 121. The two sequences share a 59.5% identity, and 74.4% similarity. 3.3% (4/121) gaps were created. Corresponding amino acids are marked "|" *(516)*. The Pairwise sequence alignment was visualized by Jalview *(488)*.